

Airbnb Price Prediction Report — Edinburgh

Generated on 2026-08-25 by the automated pipeline (`make all`). Source: InsideAirbnb.

Executive summary

This report analyzes the **Edinburgh** Airbnb market using 4,925 cleaned listings and 26 listing features. A machine-learning model was trained to predict nightly prices and then evaluated on a **holdout set** — listings it had never seen before.

The best model was **XGBoost**. On the holdout set it achieved:

- **Average error (MAE): \$68** — a typical prediction is off by about this much.
- **RMSE: \$162** — like MAE, but weighs large errors more heavily.
- **MAPE: 32%** — the average error as a share of the actual price.
- **R²: 0.64** — the share of price variation the model explains (0–1).
- **62%** of predictions were within +/-25% of the true price.

How to read this report

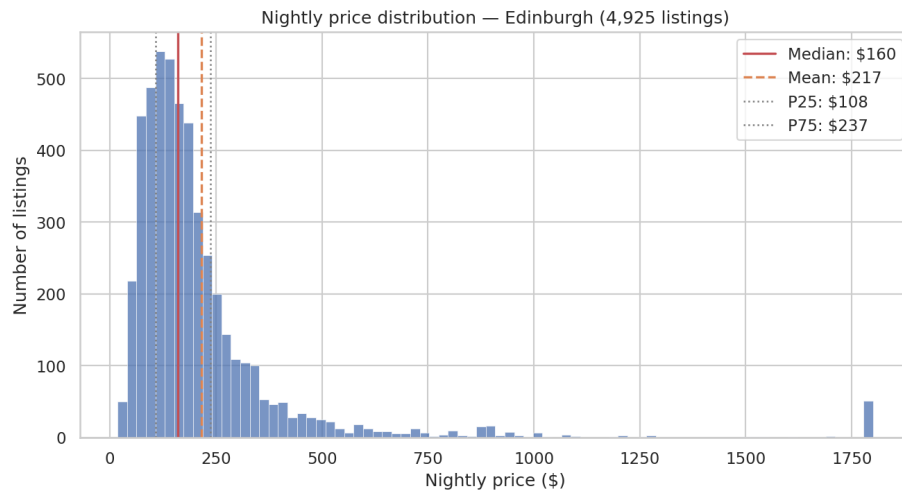
- **MAE (mean absolute error)** — the average difference between predicted and actual prices, in dollars. Lower is better; it's the easiest number to grasp ("off by about \$X on average").
- **RMSE (root mean squared error)** — similar to MAE but penalizes large mistakes more. If RMSE is much higher than MAE, a few big errors are inflating it.
- **MAPE (mean absolute percentage error)** — the same idea as MAE, but expressed as a percentage of the actual price, so it can be compared across cities.
- **R²** — how much of the variation in prices the model explains (0 = no better than guessing the average; 1 = perfect).
- **Holdout set** — the 10% of listings the model never saw while training. All accuracy numbers come from this set, so they are an honest estimate of real-world performance.
- **Top features** — the listing attributes the model leans on most when setting a price.

Data

- Source: InsideAirbnb (latest available snapshot)
- Listings after cleaning: **4,925** — median nightly price **\$160**, mean **\$217**
- Features used by the model: **26**
- Train / validation / holdout split: **3,447 / 886 / 592** listings (70/15/10, random seed 42)

Market overview

Price distribution



- Median nightly price: **\$160**; the middle 50% of listings fall between **\$108** and **\$237**. - The mean (**\$217**) sits well above the median — a minority of expensive listings pull the average up, so the **median is the better guide** to a typical price.

Price by room type

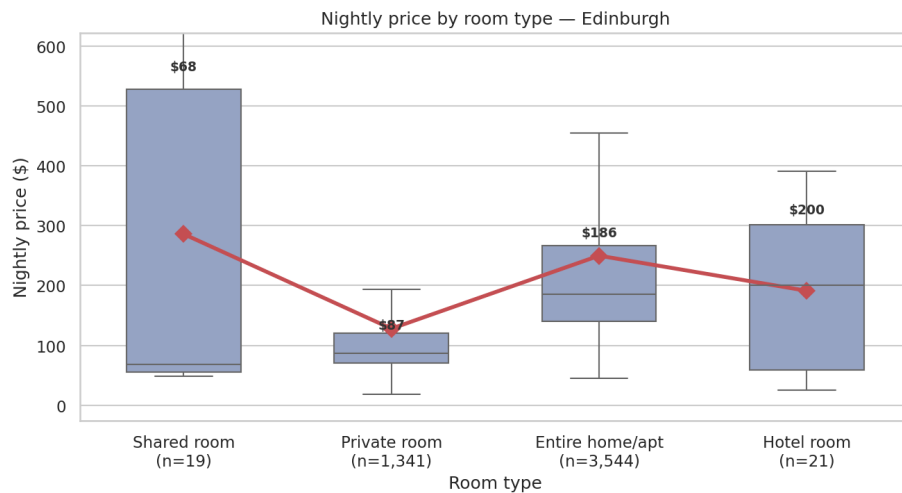


Figure 1: price_by_room_type

Room type	Listings	Median price	Mean price
Hotel room	21	\$200	\$191
Entire home/apt	3,544	\$186	\$250
Private room	1,341	\$87	\$128
Shared room	19	\$68	\$286

Price by number of bedrooms

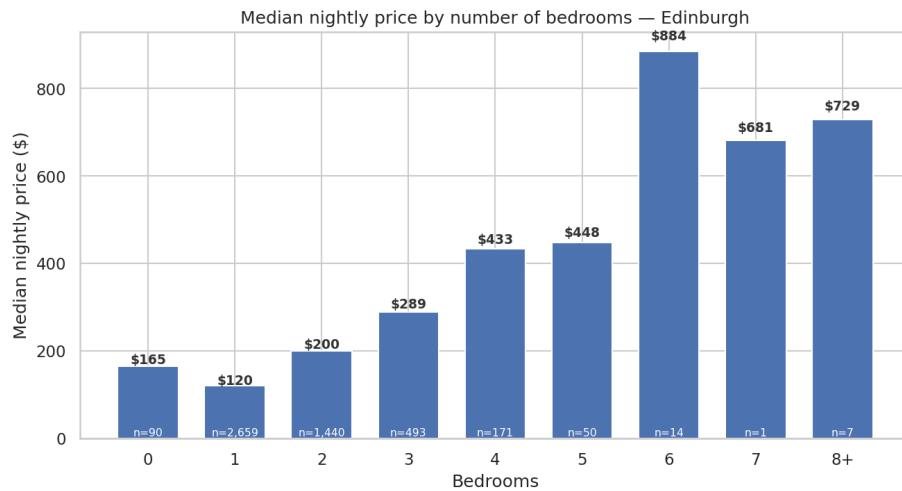


Figure 2: price_by_bedrooms

Feature correlations

Model accuracy on the holdout set

Metrics for the chosen model (**XGBoost**) on 592 unseen listings:

Metric	Value	What it means
MAE	\$68	Average prediction is off by this much
RMSE	\$162	Errors, with big misses weighted more
MAPE	32%	Average error as a % of the true price
R ²	0.64	Share of price variation explained

Accuracy in practice: 29% of predictions were within +/-10% of the true price, 62% within +/-25%, and 83% within +/-50%.

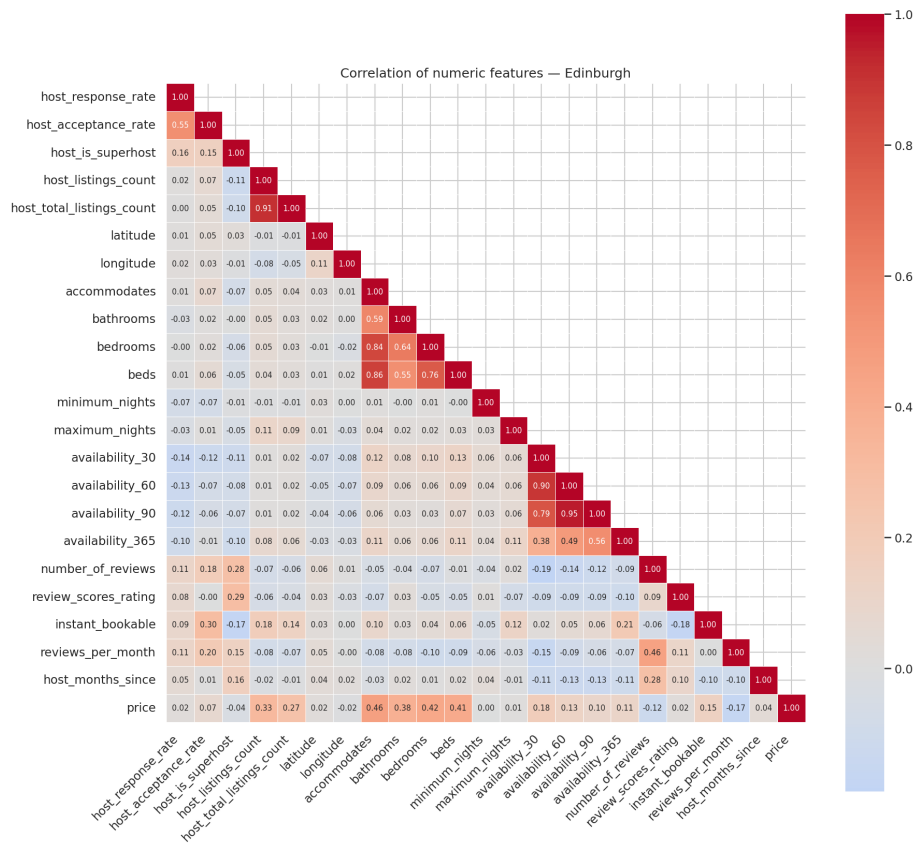
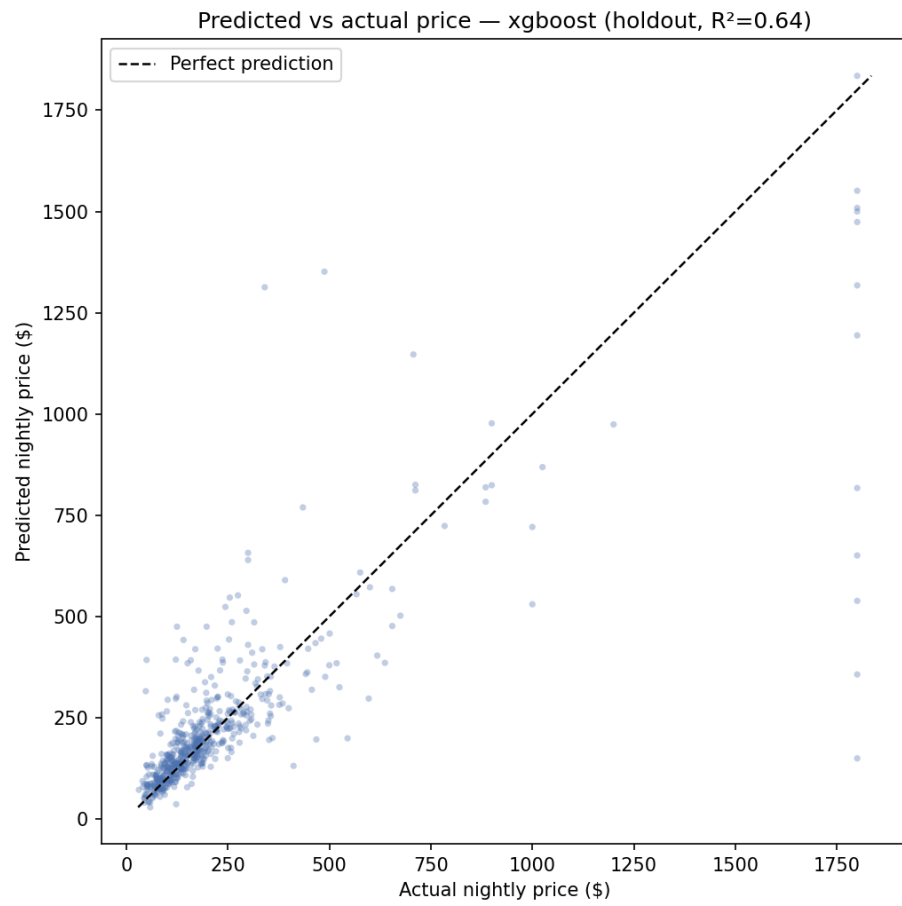
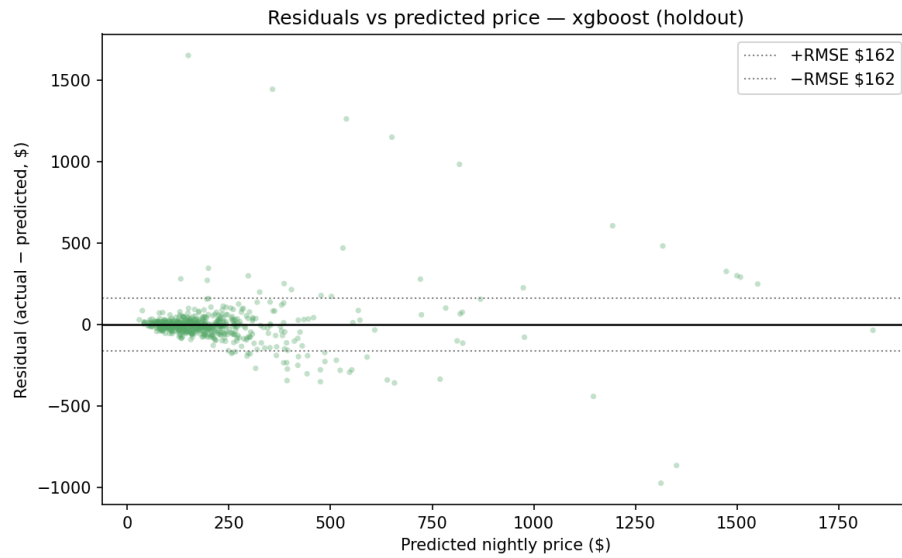


Figure 3: corr_heatmap



Points close to the dashed line are accurate predictions; points above it mean the model under-predicted.



Residual = actual - predicted. A cloud centered on the zero line means errors are balanced rather than systematically too high or too low.

Models & results

Model	Validation RMSE
Random Forest	\$137
XGBoost	\$123
TabPFN (capped sample, not directly comparable)	\$122

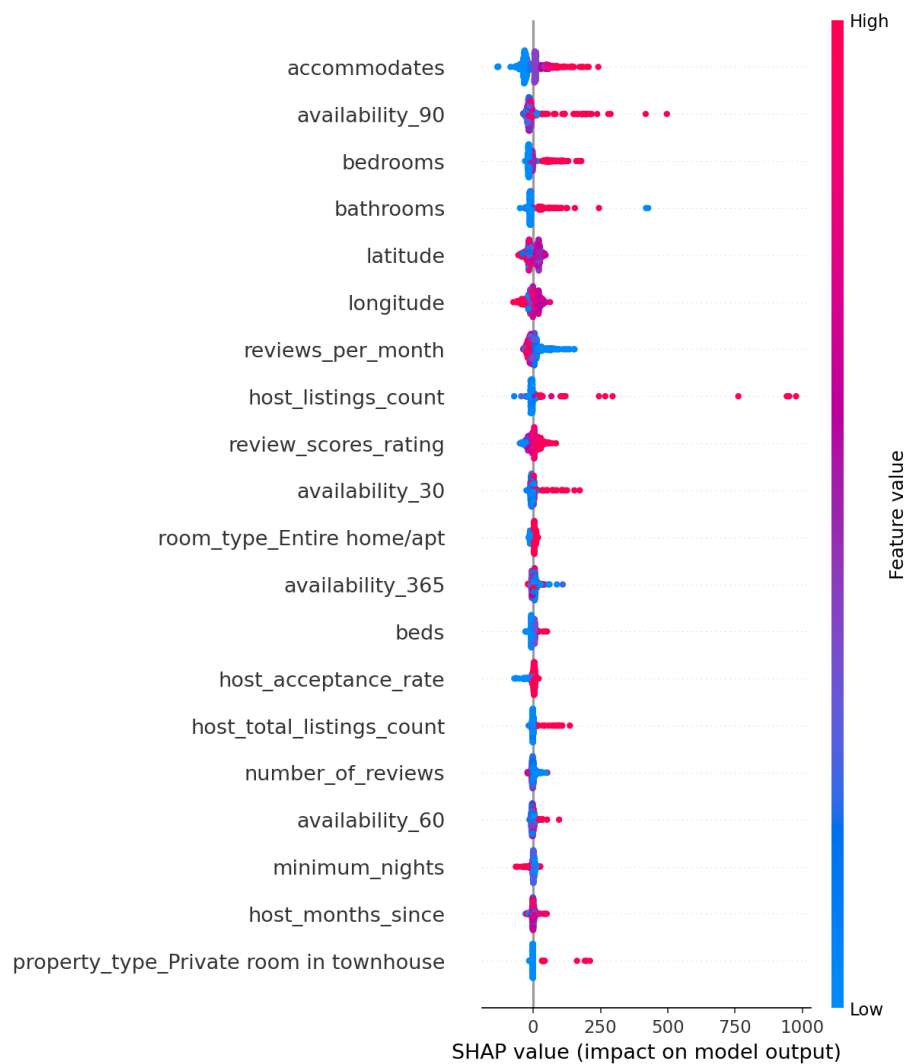
The model with the lowest validation RMSE was then scored once on the holdout set — see the accuracy section above for its final numbers.

Top features

The listing attributes that influence price the most (importance = how much the model relies on them):

Feature	Importance	Meaning
property_type_Private room in townhouse	0.083	the property type (apartment, house, etc.) (specifically Private room in townhouse)

Feature	Importance	Meaning
host_neighbourhood_New Town	0.078	the host's neighbourhood (specifically New Town)
host_neighbourhood_Cannonmills	0.071	the host's neighbourhood (specifically Cannonmills)
bedrooms	0.063	the number of bedrooms
host_neighbourhood_Fasach	0.057	the host's neighbourhood (specifically Fasach)
host_neighbourhood_Laulerie	0.042	the host's neighbourhood (specifically Laulerie)
accommodates	0.037	how many guests a listing sleeps
availability_90	0.034	availability over the next 90 days
bathrooms	0.034	the number of bathrooms
host_listings_count	0.028	how many listings the host manages



SHAP summary: each dot is one listing; colour shows the feature's value (red = high, blue = low) and position shows whether it pushed the predicted price up (right) or down (left).

Prediction examples

A few real holdout listings with the model's prediction:

Listing	Actual	Predicted	Error	Error %
accommodates= \$ 275 bedrooms=4, room_type=Entire home/apt	\$552	\$277		100.8%
accommodates= \$ 120 bedrooms=1, room_type=Private room	\$91	\$29		23.8%
accommodates= \$ 228 bedrooms=2, room_type=Entire home/apt	\$234	\$6		2.5%
accommodates= \$ 178 bedrooms=1, room_type=Entire home/apt	\$170	\$8		4.5%
accommodates= \$ 87 bedrooms=1, room_type=Private room	\$74	\$13		15.4%
accommodates= \$ 190 bedrooms=1, room_type=Entire home/apt	\$205	\$15		8.0%
accommodates= \$ 206 bedrooms=2, room_type=Entire home/apt	\$211	\$5		2.4%
accommodates= \$ 104 bedrooms=1, room_type=Private room	\$87	\$17		16.3%

Recommendations

- The best-performing model was **XGBoost** (holdout RMSE \$162, MAE \$68, R^2 0.64).
- On validation, **XGBoost** beat Random Forest by 10% on RMSE (\$123 vs \$137). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **property_type_Private room in townhouse**, **host_neighbourhood_New Town**,

host_neighbourhood_Cannonmills — i.e. `property_type_Private` room in townhouse (the property type (apartment, house, etc.) (specifically Private room in townhouse)); `host_neighbourhood_New Town` (the host's neighbourhood (specifically New Town)); `host_neighbourhood_Cannonmills` (the host's neighbourhood (specifically Cannonmills)). Hosts can use these levers when setting nightly rates.

- On average, predictions are off by about **32%** of the actual price (median error 18%), and **62%** of predictions landed within $\pm 25\%$ of the true price.
- Accuracy caveat: the holdout RMSE of \$162 is **101% of the median listing price (\$160)**, so predictions for typical listings are expected to be off by roughly $\pm \$162$.
- Next steps: add richer features (`neighbourhood_cleansed`, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Limitations

- The model only “sees” the features listed in the config — it cannot capture neighbourhood charm, amenities, photos, or seasonality.
- Extreme high-end prices are clipped at the 99th percentile during cleaning, so predictions at the very top of the market are less meaningful.
- The accuracy numbers are averages: individual predictions can be much better or much worse.
- This report is generated automatically from the pipeline artifacts and reflects the data as of 2026-08-25.

*Automated pipeline: `fetch` $\rightarrow$ `preprocess` $\rightarrow$ `EDA` $\rightarrow$ `train` $\rightarrow$ `evaluate` $\rightarrow$ `report`.
Code in `src/`, config in `configs/base.yaml`.*